New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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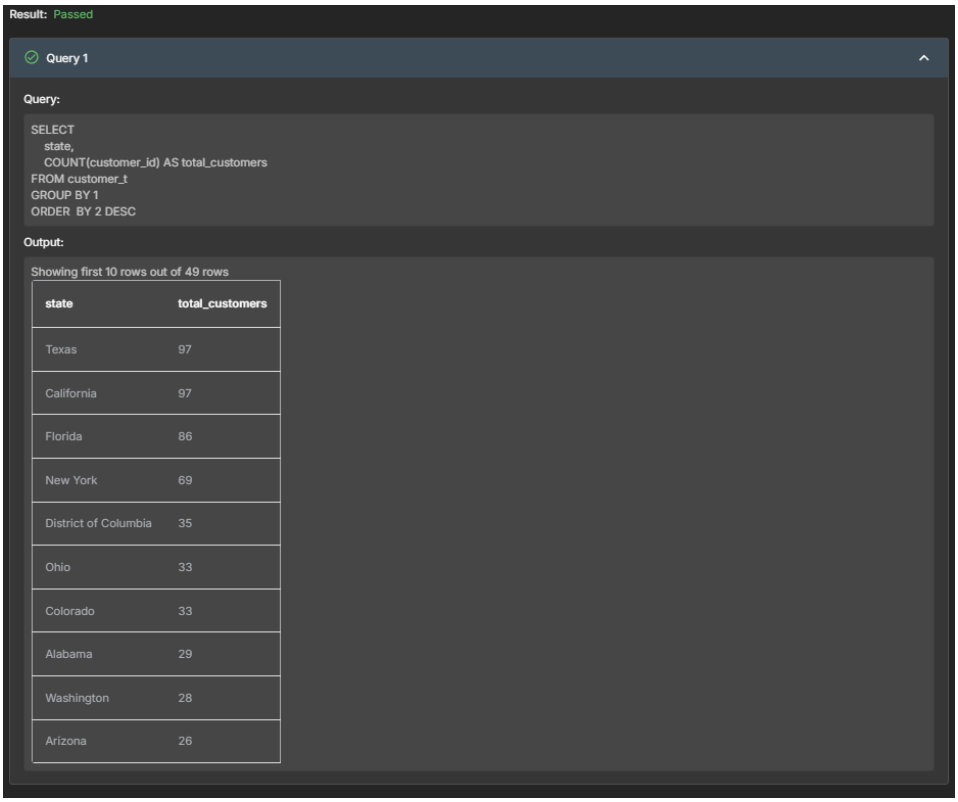
Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
   state,
   COUNT(customer_id) AS total_customers
FROM customer_t
GROUP BY 1
ORDER  BY 2 DESC;
```

Output:



Observations and Insights:

- Texas and California had the most customers with 97 in each state.
- There's a sharp decline in the total number of customers between New York and the District of Columbia
- Six of the states with the highest number of customers are in the South.

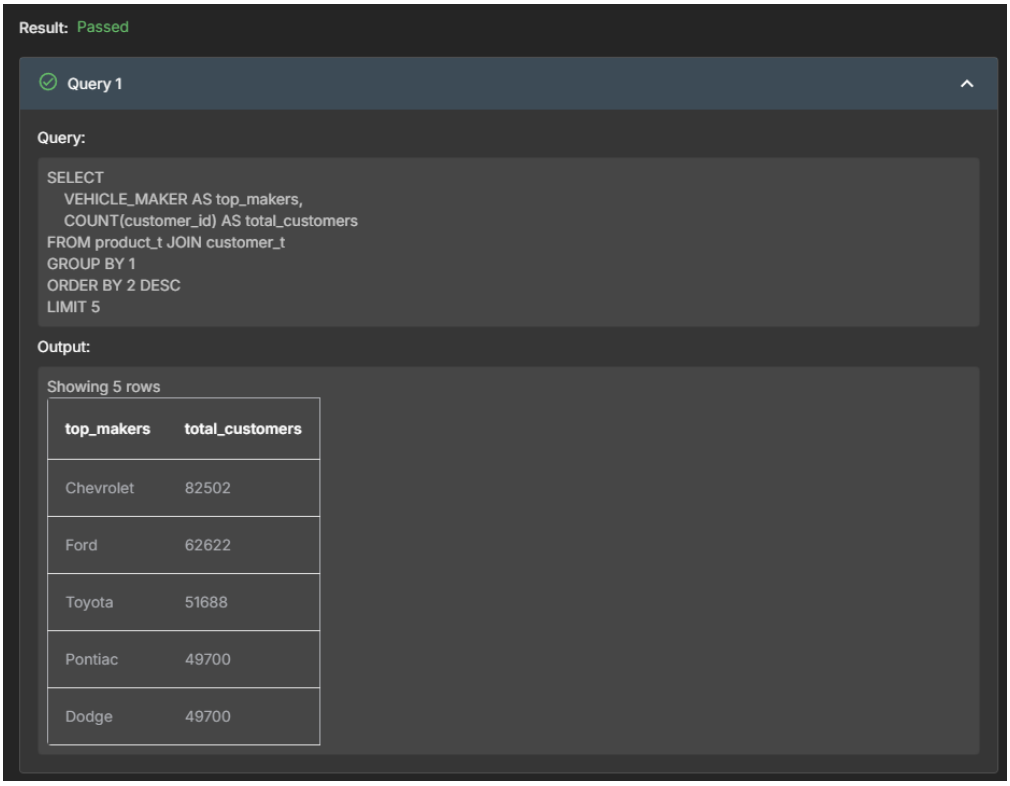
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Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
        VEHICLE_MAKER AS top_makers,
        COUNT(customer_id) AS total_customers
FROM product_t JOIN customer_t
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Output:



Observations and Insights:

- Chevrolet is the most preferred vehicle maker by the customers.
- Pontiac and Dodge are preferred by the exact same number of customers.
- Ford trails behind Chevrolet in preference with approximately 20,000 customers.

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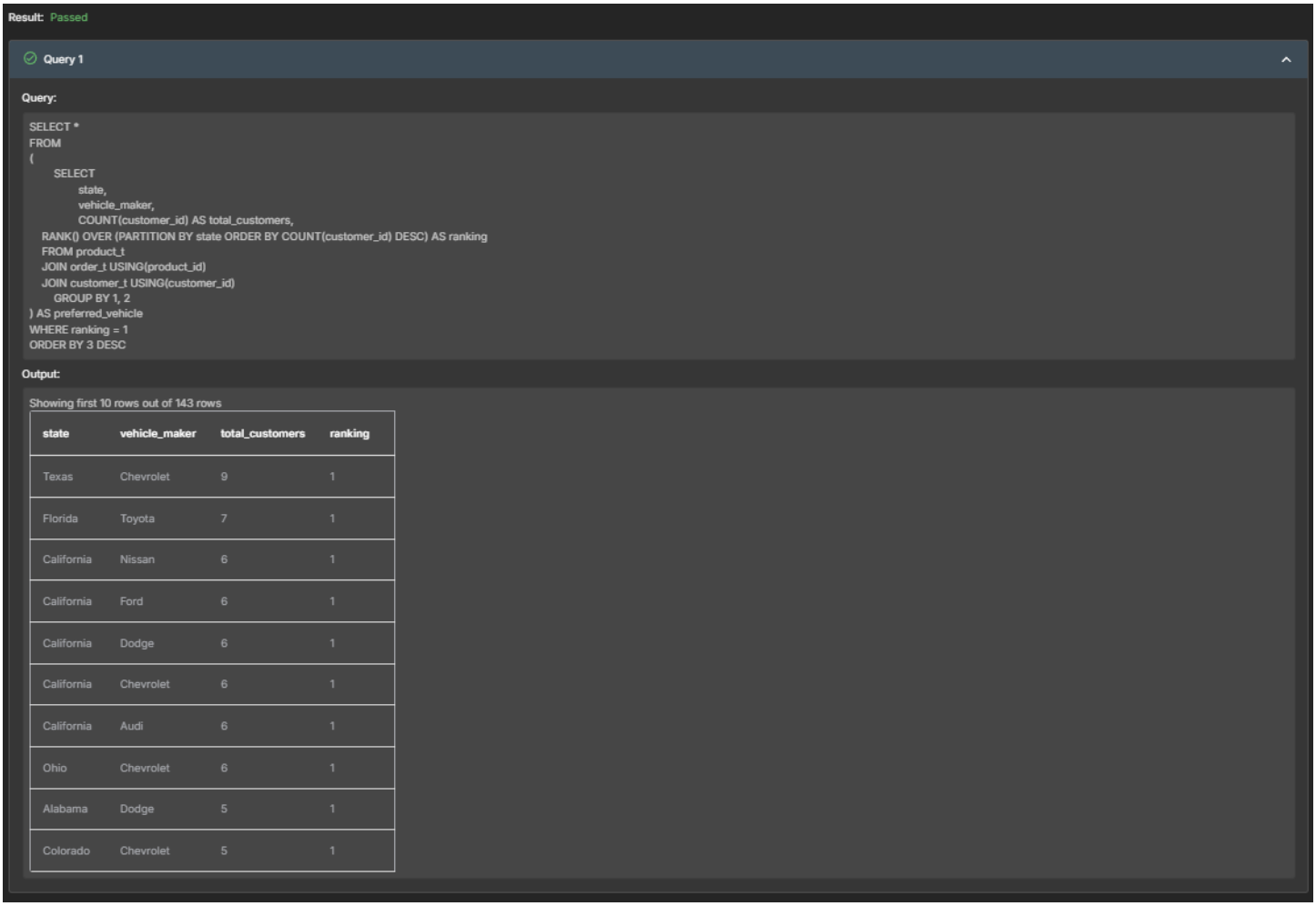
Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT *
FROM
(
   SELECT
      state,
      vehicle_maker,
      COUNT(customer_id) AS total_customers,
   RANK() OVER (PARTITION BY state ORDER BY COUNT(customer_id) DESC) AS ranking
   FROM product_t
   JOIN order_t USING(product_id)
   JOIN customer_t USING(customer_id)
   GROUP BY 1, 2
) AS preferred_vehicle
WHERE ranking = 1
ORDER BY 3 DESC;
```

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Output:



Result: Passed

Query 1

Query:

```
SELECT *
FROM
(
      SELECT
            state,
            vehicle_maker,
            COUNT(customer_id) AS total_customers,
    RANK() OVER (PARTITION BY state ORDER BY COUNT(customer_id) DESC) AS ranking
    FROM product_t
    JOIN order_t USING(product_id)
    JOIN customer_t USING(customer_id)
       GROUP BY 1, 2
) AS preferred_vehicle
WHERE ranking = 1
ORDER BY 3 DESC
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	total_customers	ranking
Texas	Chevrolet	9	1
Florida	Toyota	7	1
California	Nissan	6	1
California	Ford	6	1
California	Dodge	6	1
California	Chevrolet	6	1
California	Audi	6	1
Ohio	Chevrolet	6	1
Alabama	Dodge	5	1
Colorado	Chevrolet	5	1

Observations and Insights:

- Texas had the most decisive preferred vehicle maker with nine people preferring Chevrolet.
- Customers in California had the most split preferred vehicle maker with a whopping five makers.
- Chevrolet was the most preferred vehicle maker in five of the most responsive states.

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Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

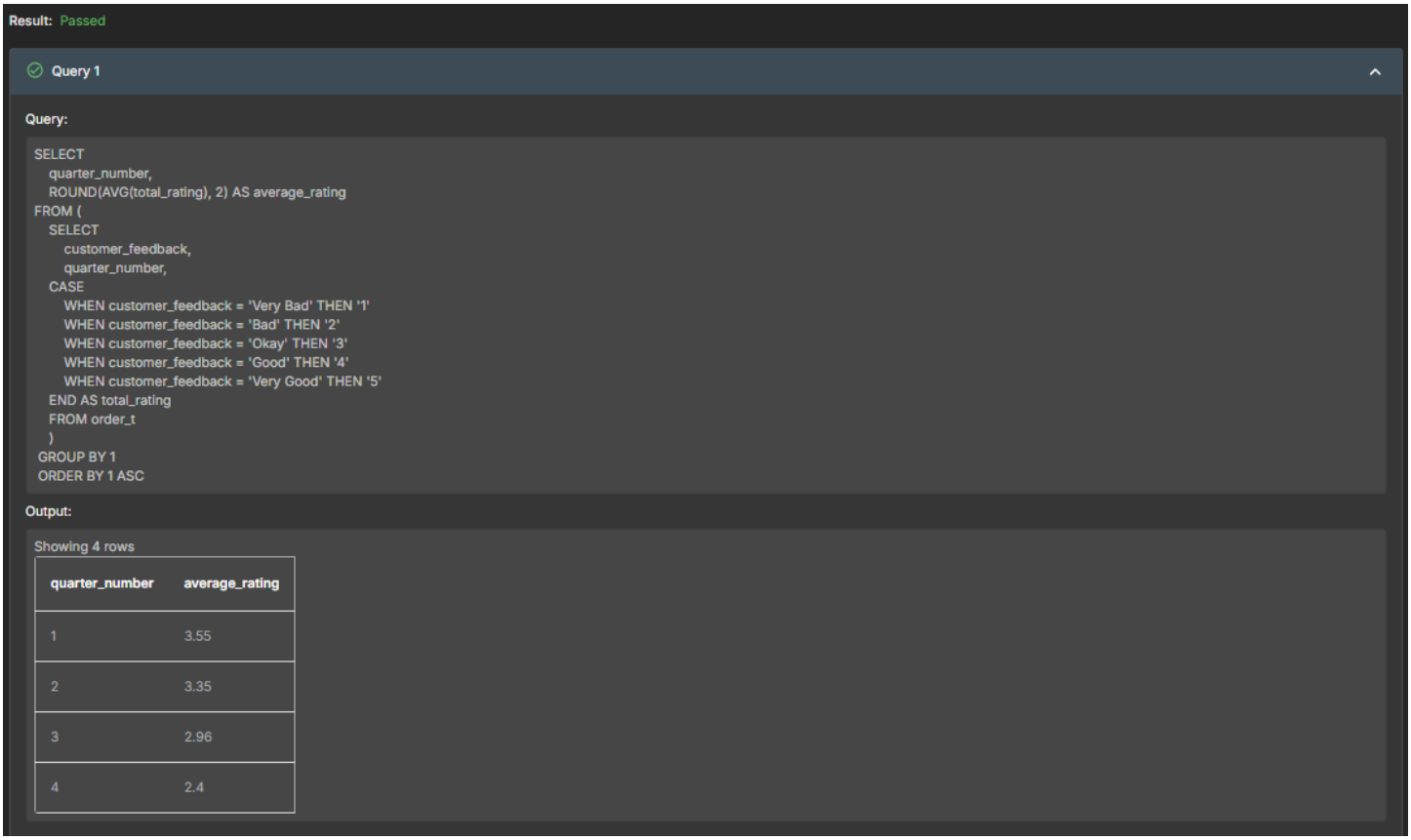
Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
  quarter_number,
  ROUND(AVG(total_rating), 2) AS average_rating
FROM (
  SELECT
    customer_feedback,
    quarter_number,
  CASE
    WHEN customer_feedback = 'Very Bad' THEN '1'
    WHEN customer_feedback = 'Bad' THEN '2'
    WHEN customer_feedback = 'Okay' THEN '3'
    WHEN customer_feedback = 'Good' THEN '4'
    WHEN customer_feedback = 'Very Good' THEN '5'
  END AS total_rating
  FROM order_t
  )
GROUP BY 1
ORDER BY 1 ASC;
```

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Output:



Observations and Insights:

- Quarter 1 had the highest average rating of 3.55%.
- The average rating has decreased throughout each quarter.
- Quarter 4 had the lowest average rating of 2.4%

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Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,
    ROUND((very_bad/total_feedback), 2) AS very_bad,
    ROUND((bad/total_feedback), 2) AS bad,
    ROUND((okay/total_feedback), 2) AS okay,
    ROUND((good/total_feedback), 2) AS good,
    ROUND((very_good/total_feedback), 2) AS very_good
FROM (
  SELECT
    quarter_number,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END), 2) AS very_bad,
    ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END), 2) AS bad,
    ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END), 2) AS okay,
    ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END), 2) AS good,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END), 2) AS very_good,
    ROUND(COUNT(customer_feedback), 2) AS total_feedback
  FROM order_t
  GROUP BY 1
  ORDER BY 1 ASC
);
```

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Output:

Result: Passed

Query 1

Query:

```
SELECT
      quarter_number,
      ROUND((very_bad/total_feedback), 2) AS very_bad,
      ROUND((bad/total_feedback), 2) AS bad,
      ROUND((okay/total_feedback), 2) AS okay,
      ROUND((good/total_feedback), 2) AS good,
      ROUND((very_good/total_feedback), 2) AS very_good
FROM (
   SELECT
      quarter_number,
      ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END), 2) AS very_bad,
      ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END), 2) AS bad,
      ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END), 2) AS okay,
      ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END), 2) AS good,
      ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END), 2) AS very_good,
      ROUND(COUNT(customer_feedback), 2) AS total_feedback
   FROM order_t
   GROUP BY 1
   ORDER BY 1 ASC
)
```

Output:

Showing 4 rows

quarter_number	very_bad	bad	okay	good	very_good
1	0.11	0.11	0.19	0.29	0.3
2	0.15	0.14	0.2	0.22	0.29
3	0.18	0.23	0.22	0.21	0.17
4	0.31	0.29	0.2	0.1	0.1

Observations and Insights:

- Highest decrease of “Very Good” ratings was between quarters 2 and 3.
- Highest increase of “Very Bad” ratings was between quarters 3 and 4.
- Customers have become increasingly dissatisfied throughout each quarter.

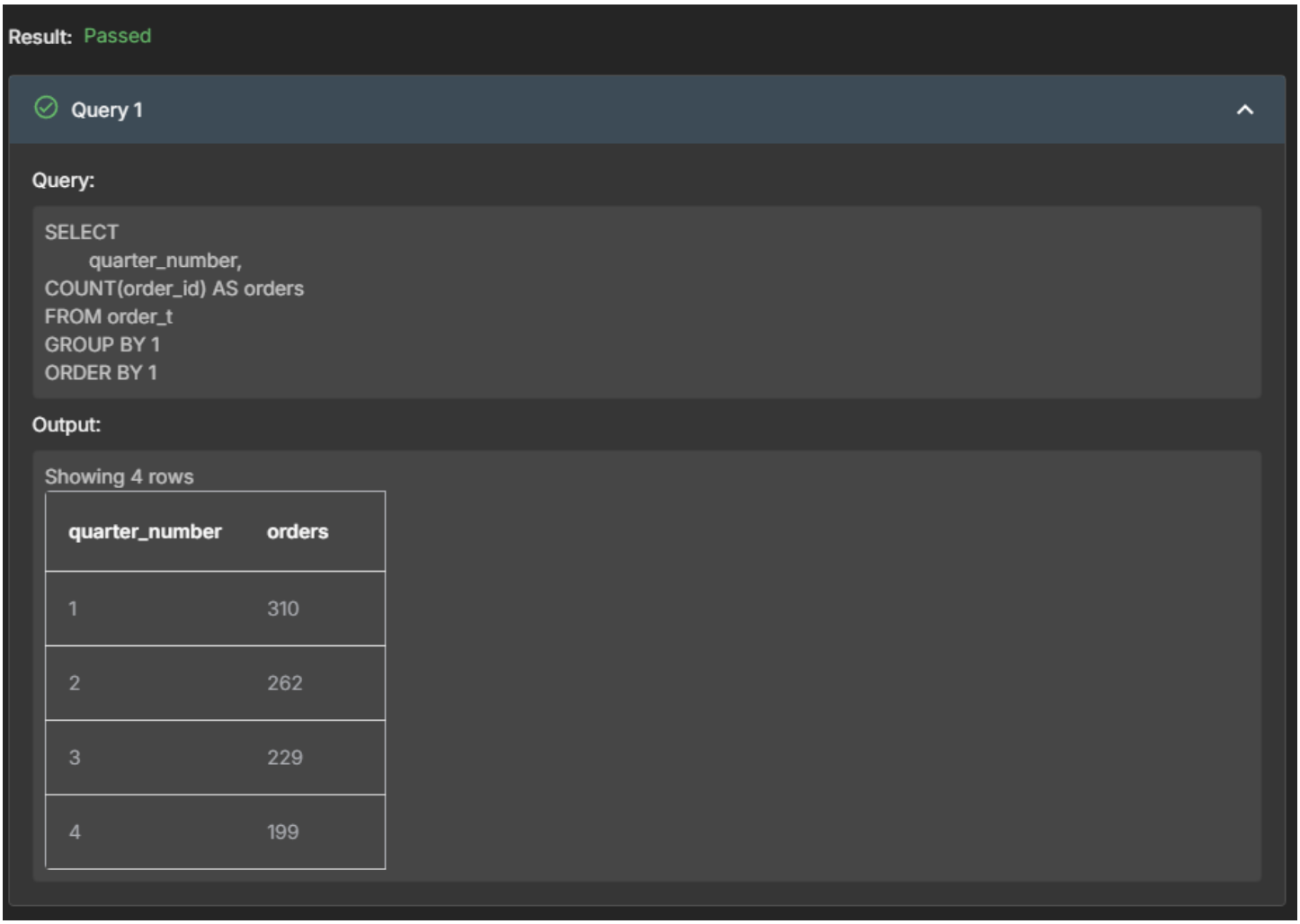
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Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
   quarter_number,
   COUNT(order_id) AS orders
FROM order_t
GROUP BY 1
ORDER BY 1;
```

Output:



Observations and Insights:

- The number of orders steadily decreases by each quarter.
- The first quarter had the highest number of orders with 310.
- The fourth quarter had the lowest number of orders with 199.

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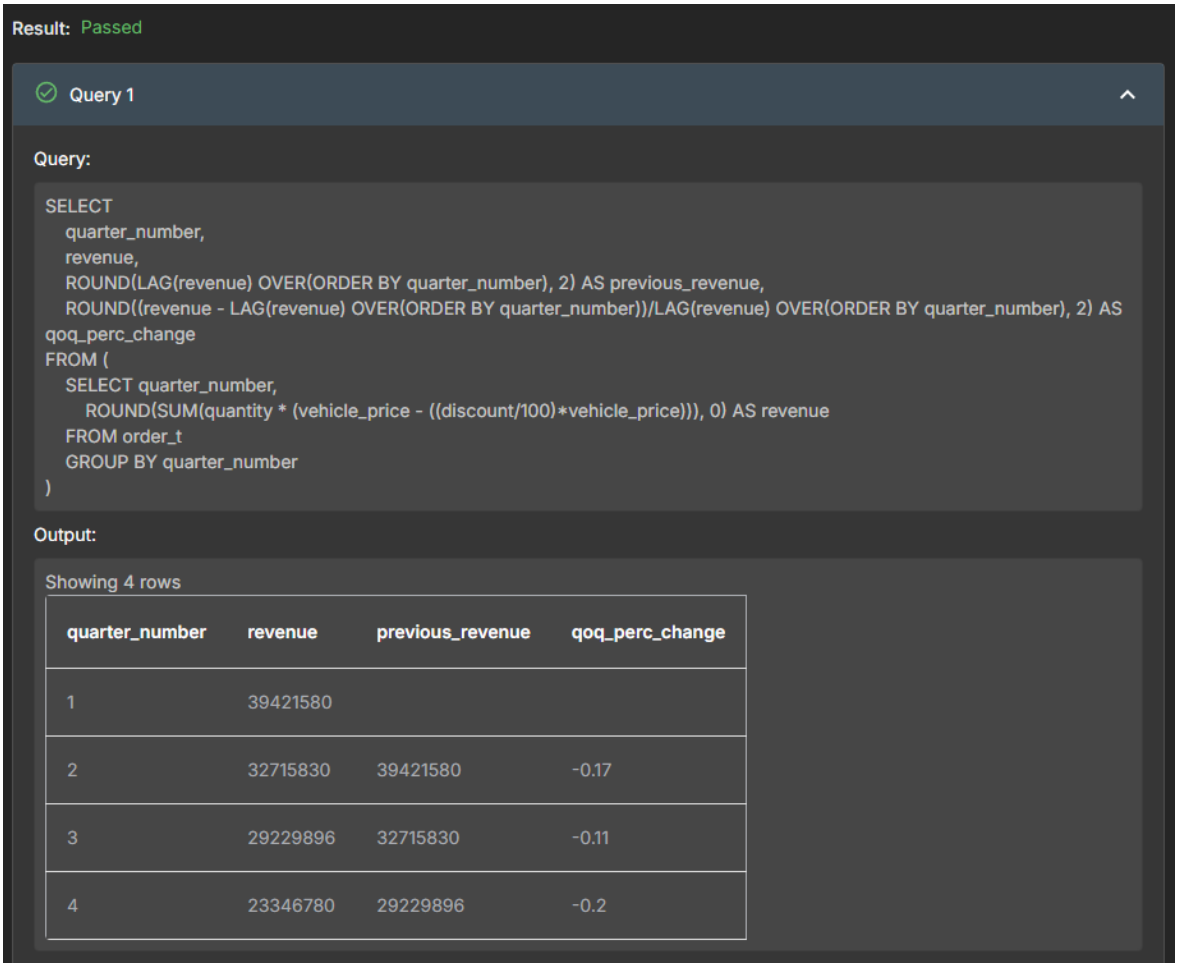
Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
   quarter_number,
   revenue,
   ROUND(LAG(revenue) OVER(ORDER BY quarter_number), 2) AS previous_revenue,
   ROUND((revenue - LAG(revenue) OVER(ORDER BY quarter_number))/LAG(revenue) OVER(ORDER BY quarter_number), 2) AS qoq_perc_change
FROM (
   SELECT quarter_number,
      ROUND(SUM(quantity * (vehicle_price - ((discount/100)*vehicle_price))), 0) AS revenue
   FROM order_t
   GROUP BY quarter_number
);
```

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Output:



Result: Passed

Query 1

Query:

```
SELECT
   quarter_number,
   revenue,
   ROUND(LAG(revenue) OVER(ORDER BY quarter_number), 2) AS previous_revenue,
   ROUND((revenue - LAG(revenue) OVER(ORDER BY quarter_number))/LAG(revenue) OVER(ORDER BY quarter_number), 2) AS
qoq_perc_change
FROM (
   SELECT quarter_number,
      ROUND(SUM(quantity * (vehicle_price - ((discount/100)*vehicle_price))), 0) AS revenue
   FROM order_t
   GROUP BY quarter_number
)
```

Output:

Showing 4 rows

quarter_number	revenue	previous_revenue	qoq_perc_change
1	39421580		
2	32715830	39421580	-0.17
3	29229896	32715830	-0.11
4	23346780	29229896	-0.2

Observations and Insights:

- The revenue decreased within each quarter.
- The quarter-on-quarter percentage change decreased from each quarter.
- The average percentage change is approximately 16%.

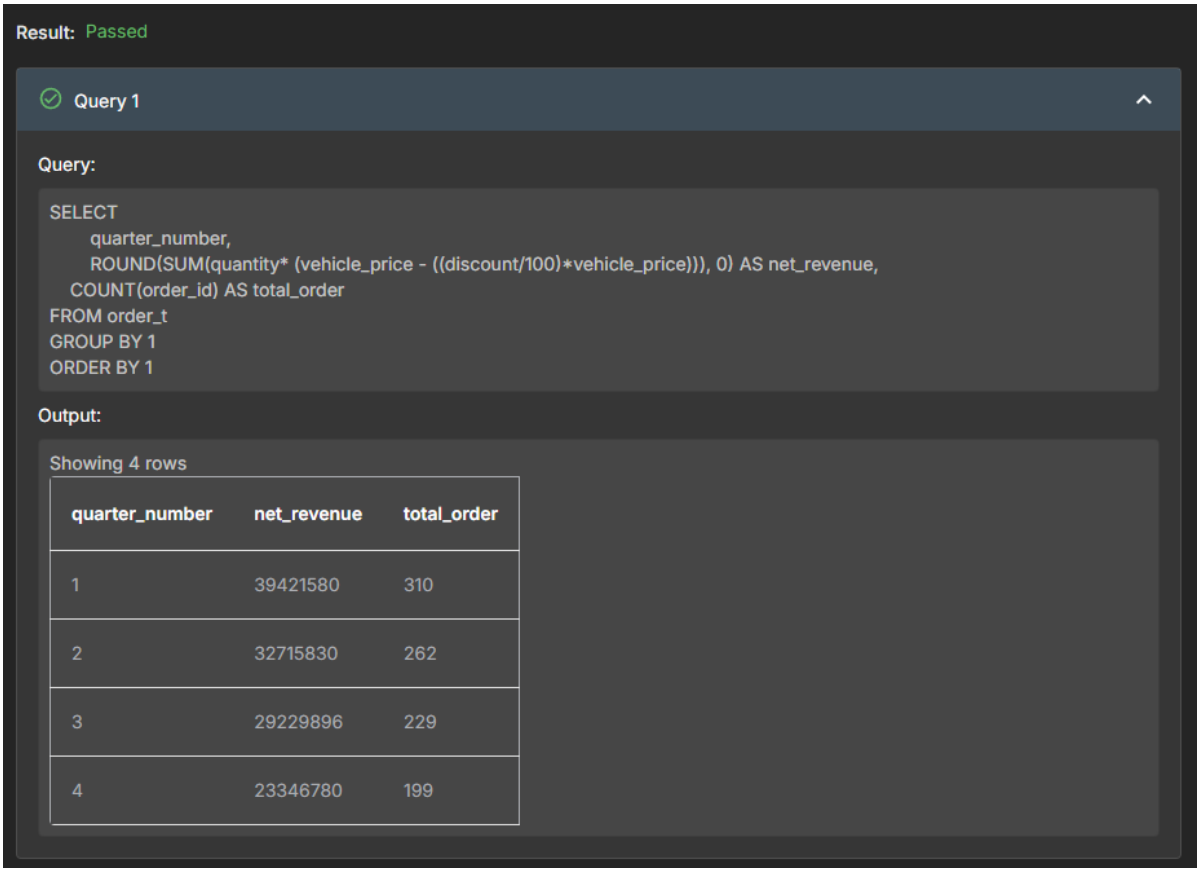
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Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
        quarter_number,
        ROUND(SUM(quantity * vehicle_price), 0) AS net_revenue,
        COUNT(order_id) AS total_orders
FROM order_t
GROUP BY 1
ORDER BY 1;
```

Output:



Observations and Insights:

- The number of orders has decreased in conjunction with the net revenue.
- The most orders were placed in the first quarter.
- The least amount of orders were placed in the fourth quarter.

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Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
   credit_card_type,
   ROUND(AVG(discount), 2) AS average_discount
FROM order_t t1
INNER JOIN customer_t t2
 ON t1.customer_id = t2.customer_id
GROUP BY 1
ORDER BY 2 DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
      credit_card_type,
      ROUND(AVG(discount), 2) AS average_discount
FROM order_t t1
INNER JOIN customer_t t2
 ON t1.customer_id = t2.customer_id
GROUP BY 1
ORDER BY 2 DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61
jcb	0.61
diners-club-us-ca	0.61

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Observations and Insights:

- Laser credit card users had the highest average discount offered.
- Some of the highest average discounts offered were on five different credit cards.
- Credit card discounts are firmly around .60

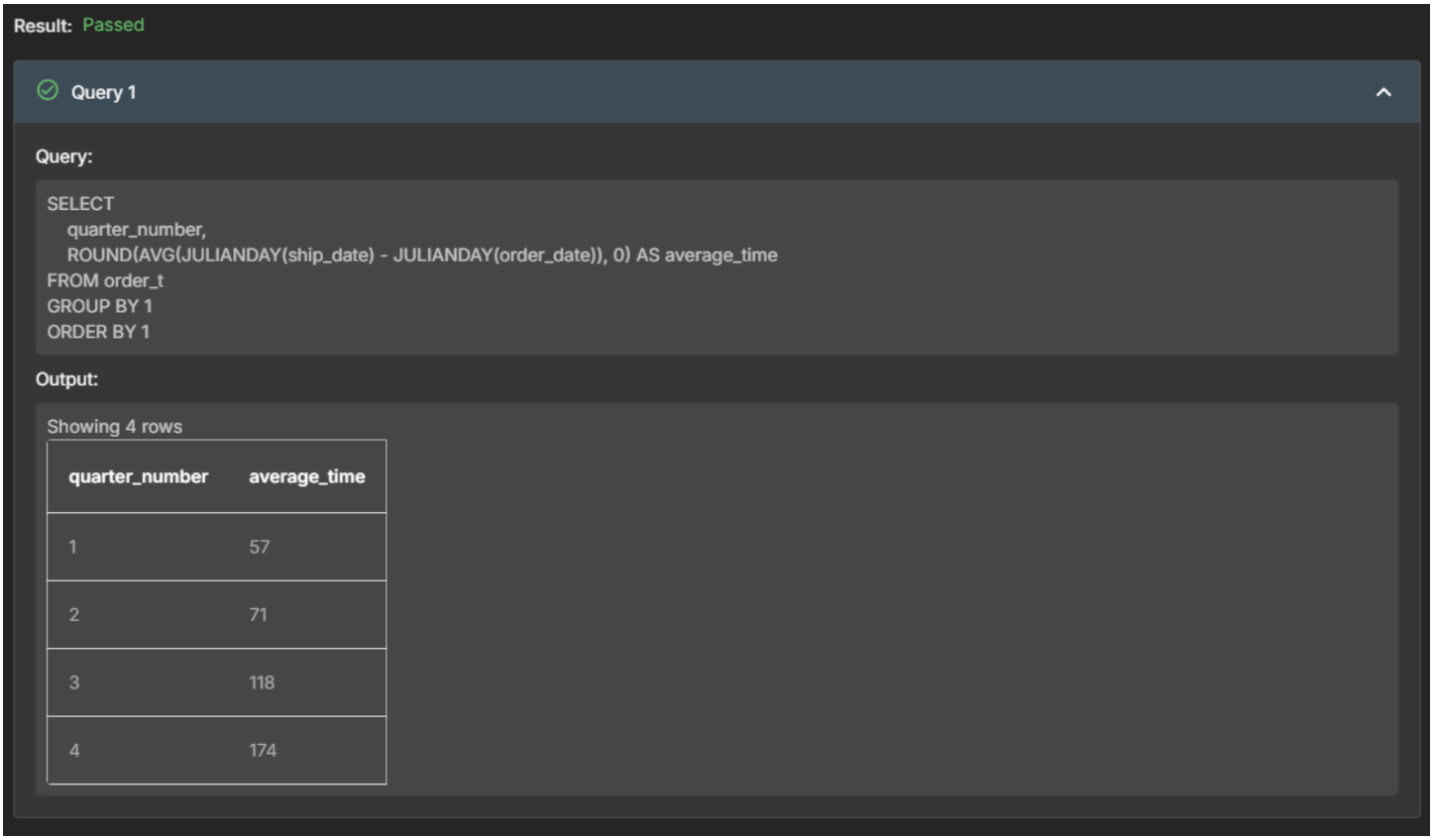
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Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   quarter_number,
   ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 0) AS average_days
FROM order_t
GROUP BY 1
ORDER BY 1;
```

Output:



Observations and Insights:

- The average time taken to ship the orders has increased each quarter.
- The shortest average time per order is in the first quarter, with 57 days.

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- The longest average time per order is in the fourth quarter with 174 days.

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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
<$124.6M>	<1,000 Orders>	<994 Customers>	<3.06/5>
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
<$23.3M>	<199 Orders>	<174 Days>	<20%>

Business Recommendations

- Focus on customer feedback and how to fix commonly recurring problems.
- Make a course of action plan on how to win back over customers.
- Look into how employee morale could be improved.

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